

# INFLATION

Fernando M. Martin  
FRB St. Louis

Updated May 19, 2026

# THE (ELUSIVE) ROAD TOWARDS 2% INFLATION

Annualized PCE inflation through the year in March 2026:  
5.6% headline, 4.2% excluding energy and 4.4% Core.

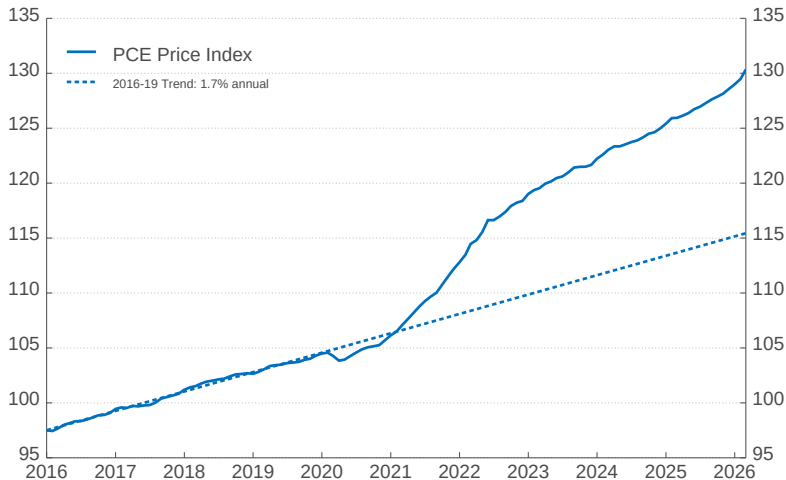
Inflation rates mostly unchanged for 2023, 2024 and 2025.

Notable increase in prices of core goods and core services ex. housing.

# Prices and Inflation

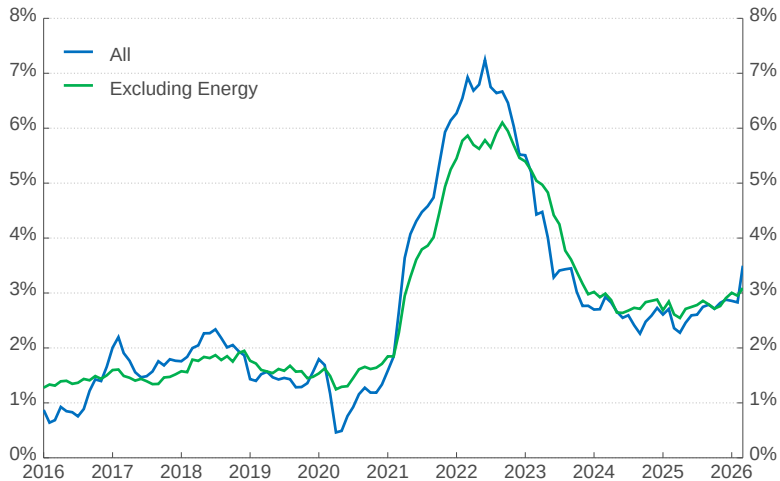
# PRICES: STEADY INCREASE

INITIAL DECLINE DURING PANDEMIC—FAST INCREASE AFTERWARDS—ABOVE-TARGET GROWTH RATE SINCE 2023



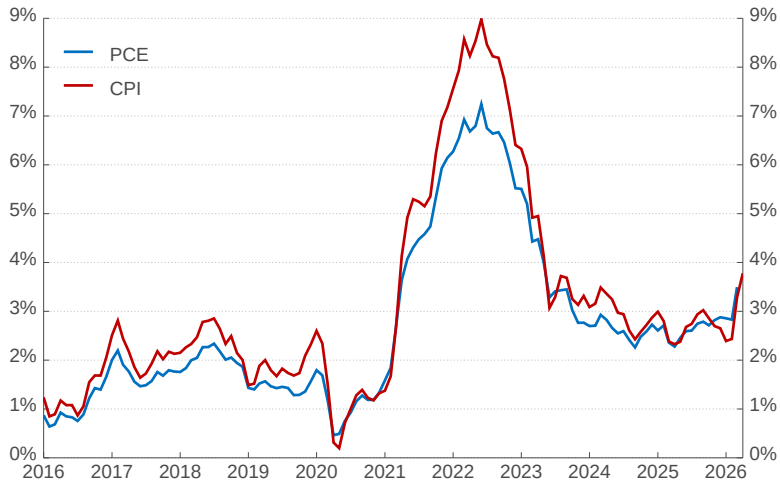
# INFLATION: FAST DECLINE IN 2023, STALLING IN 2024-25

INFLATION HAS REMAINED ABOVE 2% SINCE MARCH 2021—DYNAMICS AMPLIFIED BY ENERGY ▷ HISTORICAL SERIES



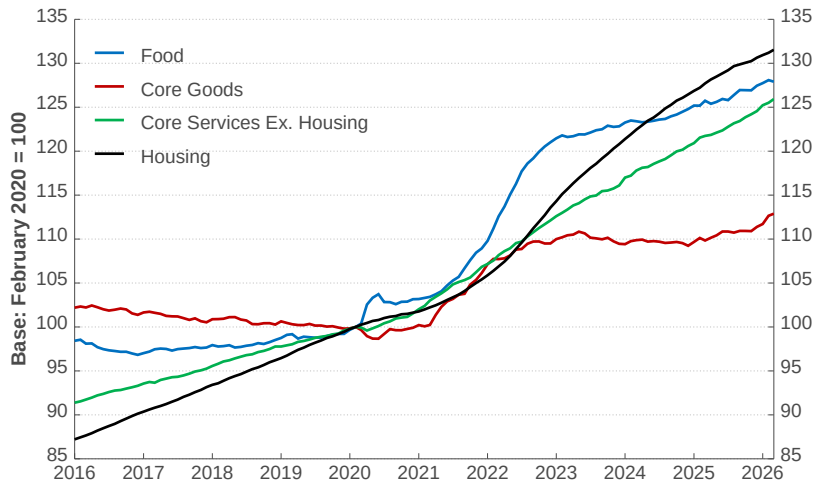
# COMPARING DYNAMICS IN PCE AND CPI

LACK OF SUBSTANTIAL PROGRESS TOWARDS TARGET, WITH SOME RECENT DIVERGENCE ▷ EXCLUDING ENERGY



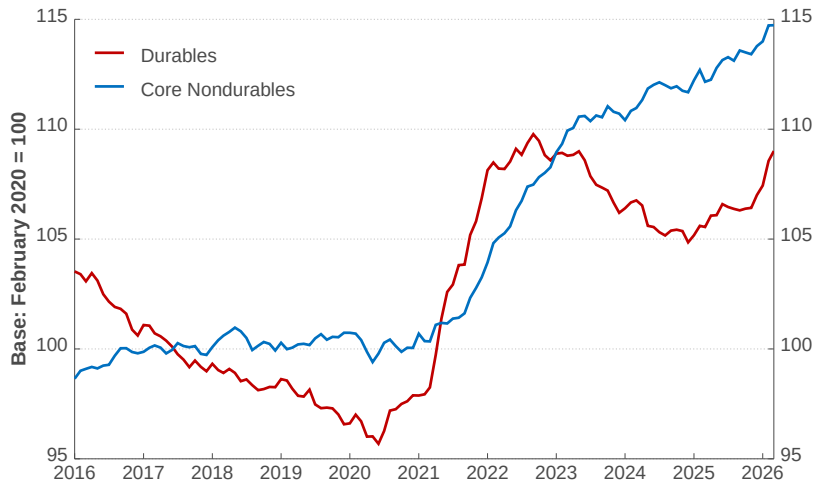
# COMPONENTS OF PCE PRICE INDEX

EFFECT OF TARIFFS EVIDENT ON FOOD AND CORE GOODS PRICES



# CORE GOODS: DURABLES VS. NONDURABLES

EFFECT OF TARIFFS EVIDENT ON PRICES OF CORE DURABLE GOODS—CORE NONDURABLES GROWING STEADILY





# INFLATION REMAINS ABOVE THE FED'S TARGET

AGGREGATE INFLATION MEASURES—STRONG MODERATION IN 2023 BUT LACK OF PROGRESS IN 2024-25 ▷ CPI

## Annualized PCE Inflation Rates

|          | All Goods<br>& Services | Excluding<br>Energy | Excluding Food<br>& Energy (Core) |
|----------|-------------------------|---------------------|-----------------------------------|
| 2016–19  | 1.7%                    | 1.6%                | 1.7%                              |
| 2020     | 1.3%                    | 1.7%                | 1.5%                              |
| 2021     | 6.1%                    | 5.2%                | 5.2%                              |
| 2022     | 5.5%                    | 5.5%                | 5.0%                              |
| 2023     | 2.8%                    | 3.0%                | 3.1%                              |
| 2024     | 2.7%                    | 2.9%                | 3.0%                              |
| 2025     | 2.9%                    | 2.9%                | 3.0%                              |
| 2026 YTD | 5.6%                    | 4.2%                | 4.4%                              |

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through March.

# PANDEMIC DISRUPTED SOME PRE-EXISTING INFLATION TRENDS

STRONG REVERSALS IN FOOD, ENERGY AND CORE GOODS; MODERATION IN CORE SERVICES AND HOUSING ▸ CPI

## Annualized PCE Inflation Rates by Categories

|          | Food  | Energy | Core Goods | Core Services<br>ex. Housing | Housing |
|----------|-------|--------|------------|------------------------------|---------|
| 2016–19  | 0.2%  | 4.2%   | −0.6%      | 2.2%                         | 3.4%    |
| 2020     | 4.0%  | −7.5%  | 0.1%       | 2.0%                         | 2.2%    |
| 2021     | 5.6%  | 30.3%  | 6.2%       | 5.2%                         | 3.7%    |
| 2022     | 11.1% | 6.7%   | 3.2%       | 4.9%                         | 7.8%    |
| 2023     | 1.5%  | −2.1%  | 0.0%       | 3.5%                         | 6.4%    |
| 2024     | 1.6%  | −1.0%  | −0.2%      | 3.9%                         | 4.7%    |
| 2025     | 2.1%  | 2.2%   | 2.0%       | 3.3%                         | 3.3%    |
| 2026 YTD | 1.4%  | 48.6%  | 5.5%       | 4.5%                         | 2.8%    |

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through March.

# ESTIMATING IMPACT OF TARIFFS ON INFLATION

SIMPLE BACK-OF-THE-ENVELOPE CALCULATION VS. SOPHISTICATED MODELS.

Annual PCE inflation in 2025: 2.9%.

Compute inflation ex tariffs by using inflation rates from one year ago for food and core goods.

- ▶ Tariffs contributed 0.50 percentage points to inflation.
- ▶ Inflation excluding tariffs in 2025 estimated at 2.4%.
- ▶ Other indexes ex tariffs: Ex Energy at 2.4% and Core at 2.5%.

Lower estimates of impact when assuming core nondurables were not affected by tariffs: 0.40.

Thus, estimated inflation ex tariffs in 2025 was 2.4-2.5%.

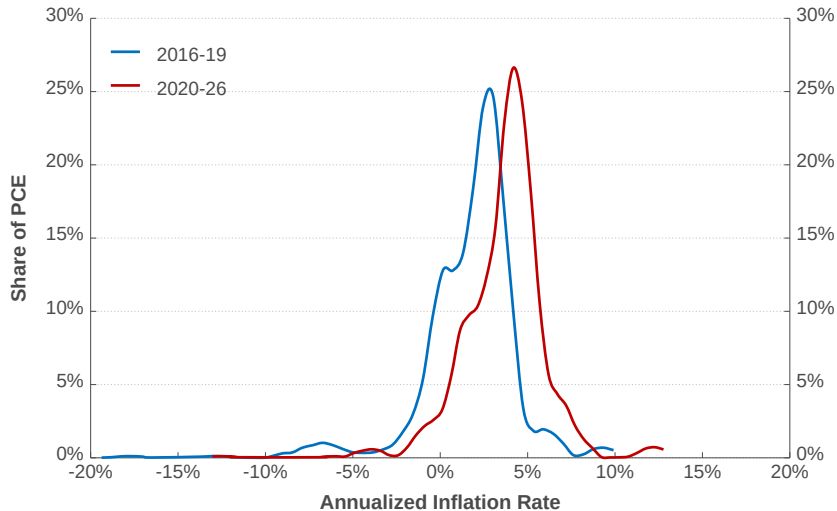
Model by STL FRB staff yields similar estimates of impact: 0.50 (headline) and 0.38 (Core).

# Inflation Distributions

# HIGH INFLATION EPISODE WAS WIDESPREAD

► METHODOLOGY

ESTIMATED DISTRIBUTION OF PCE INFLATION ► CORE



# INFLATION DISTRIBUTIONS: PROGRESSION FROM 2016-19 TO 2026

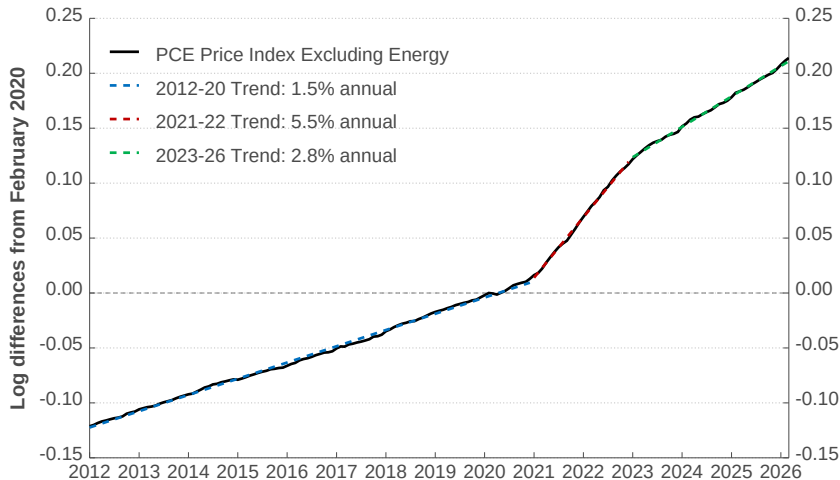
ESTIMATED DISTRIBUTION OF PCE INFLATION

# Inflation Regimes

# PRICE LEVEL DYNAMICS: THREE REGIMES

[► BLOG](#)

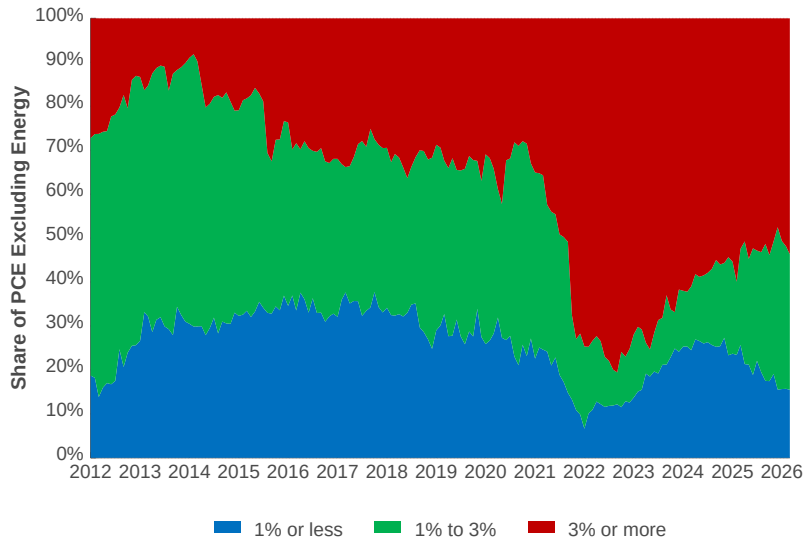
(1) BELOW-TARGET INFLATION: 2012-20 (2) GREAT INFLATION: 2021-22 (3) ABOVE-TARGET INFLATION: 2023-25





# EXPENDITURE SHARES BY INFLATION RATES

ALMOST HALF OF CONSUMPTION EXPENDITURES ARE ON PRODUCTS WITH INFLATION RATES OF 3% OR MORE



# LONG-TERM INFLATION EXPECTATIONS HAVE RISEN

MARKETS VS. MODELS VS. FORECASTERS

## 10-Year Inflation Expectations — Period Averages

|                           | TIPS | Swaps | Cleveland | DKW  | SPF  |
|---------------------------|------|-------|-----------|------|------|
| 2012–20                   | 1.9% | 2.2%  | 1.7%      | 2.2% | 2.0% |
| 2021–22                   | 2.4% | 2.7%  | 1.9%      | 2.2% | 2.3% |
| 2023–26                   | 2.3% | 2.5%  | 2.3%      | 2.6% | 2.2% |
| Change 2012-20 to 2023-26 | 0.4% | 0.3%  | 0.5%      | 0.4% | 0.1% |

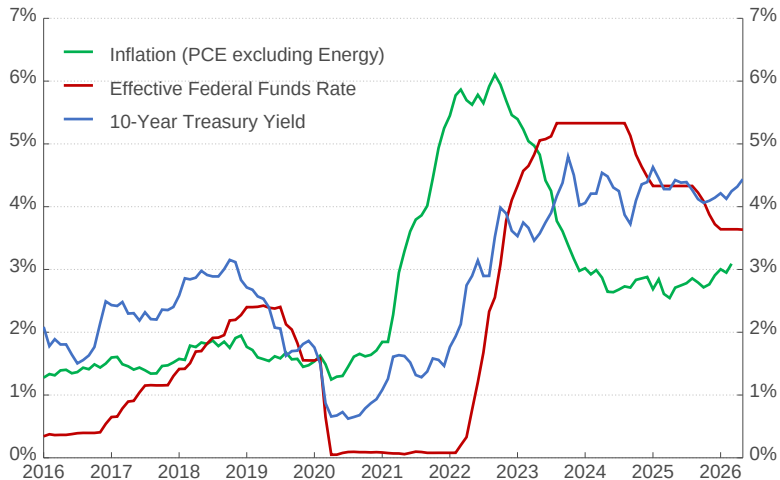
SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is through April.

# Interest Rates

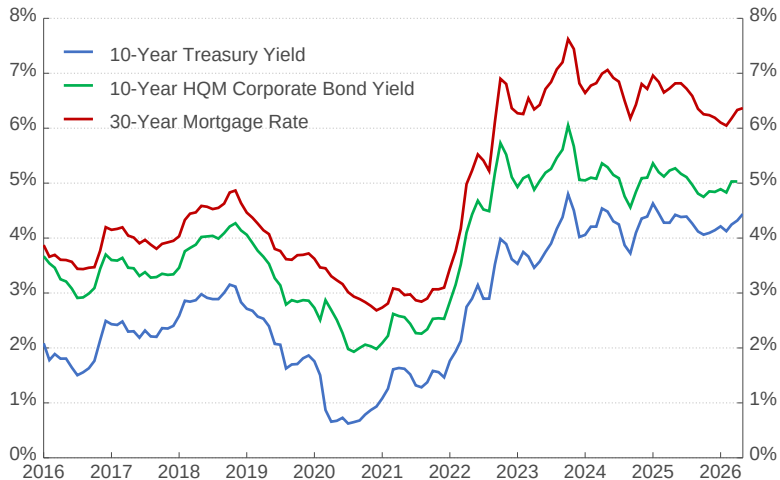
# MONETARY POLICY

EASING CYCLES IN SEPTEMBER-DECEMBER 2024 AND SEPTEMBER-DECEMBER 2025



# MONETARY POLICY: IMPACT ON LONG-TERM RATES

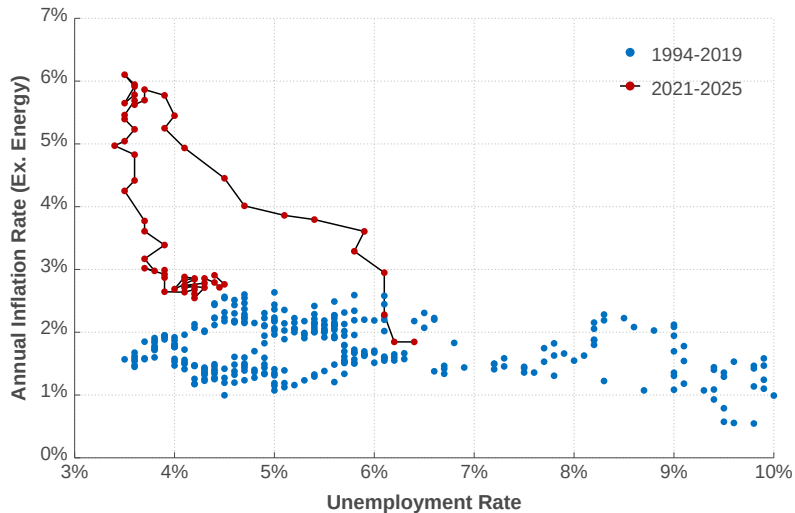
LONG-TERM RATES HAVE NOT YET RESPONDED TO EASING CYCLE—BUT RECENT DECLINE IN MORTGAGES



# Phillips Curve

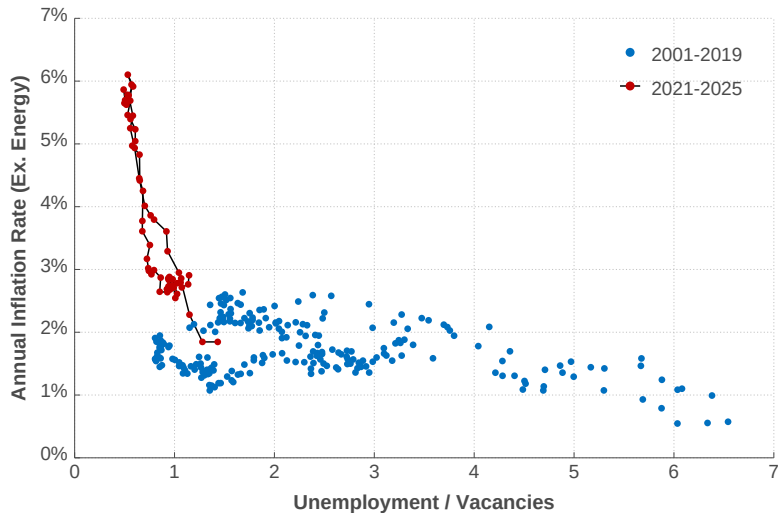
# THE LABOR MARKET REMAINS STABLE

THE INFLATION-UNEMPLOYMENT TRADE-OFF (PHILLIPS CURVE): THE CURVE HAS BECOME A DOT



# INFLATION VS. LABOR MARKET TIGHTNESS

UNEMPLOYMENT-TO-VACANCY RATIO DROPPED BELOW UNITY

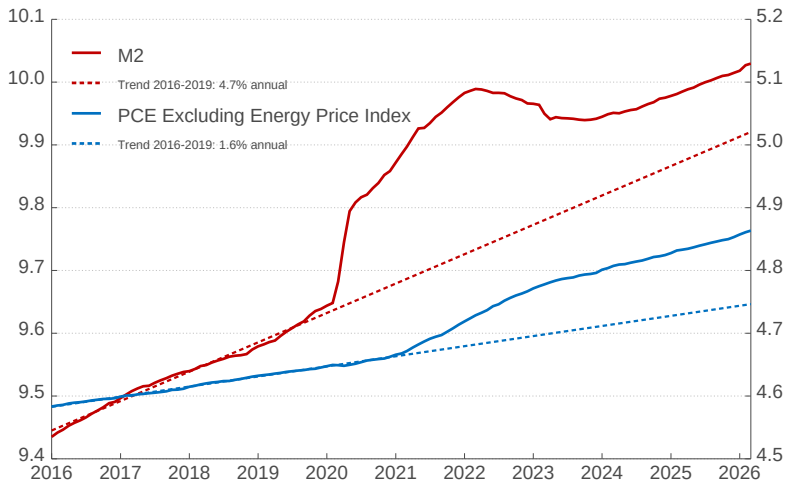




$$\mathbf{M} \times \mathbf{V} = \mathbf{P} \times \mathbf{Y}$$

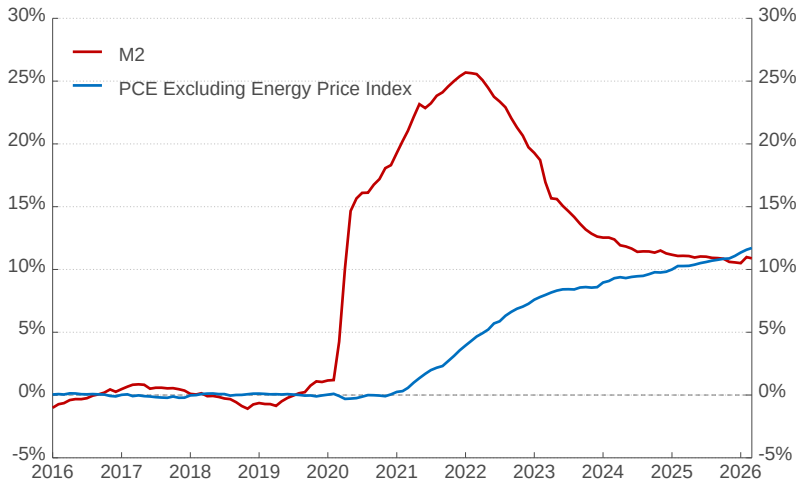
# MONEY VS PRICES: MONETARY EXPANSION DURING COVID

IN LOGS: M2 ON LEFT AXIS, PRICES ON RIGHT AXIS—M2 GREW AT SAME RATE AS DISPOSABLE INCOME PRE-COVID



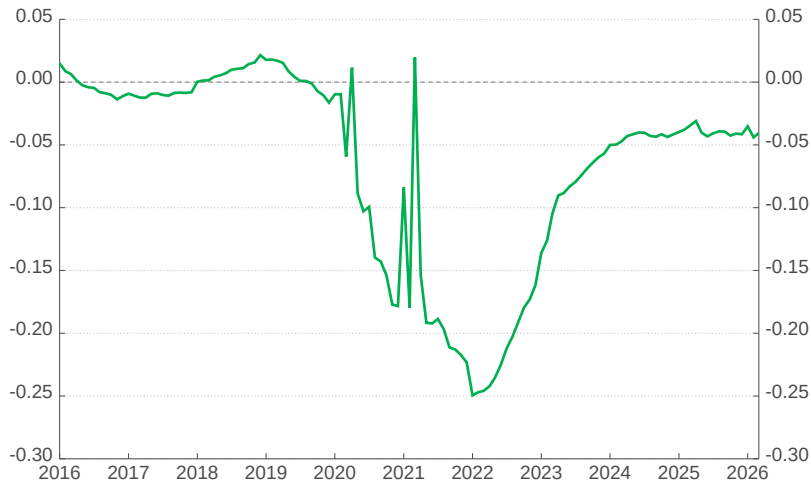
# PRICE INCREASE ONLY SLIGHTLY BELOW MONETARY EXPANSION

DEVIATIONS FROM PRE-COVID TREND



# STABLE VELOCITY IN 2024-25—SLIGHTLY BELOW PRE-PANDEMIC

$\text{LN VELOCITY} = \text{LN M2} - \text{LN DISPOSABLE PERSONAL INCOME}$  (BASE: 2016-19 = 0)



# CONCLUSIONS

In 2025, food, energy and core goods contributing positively to inflation; core services and housing inflation trending downwards.

Trends in prices:

- ▶ food and durable goods show the impact of tariffs
- ▶ nondurables and core services ex. housing have kept a steady growth pace
- ▶ housing significantly moderating its pace, especially since September

Inflation (excluding energy) remains in above-target regime:

- ▶ 2016-19: 1.5%
- ▶ 2021-22: 5.5%
- ▶ 2023-25: 2.8%

Inflation ex tariffs in 2025 estimated at 2.4-2.5%.

# Appendix

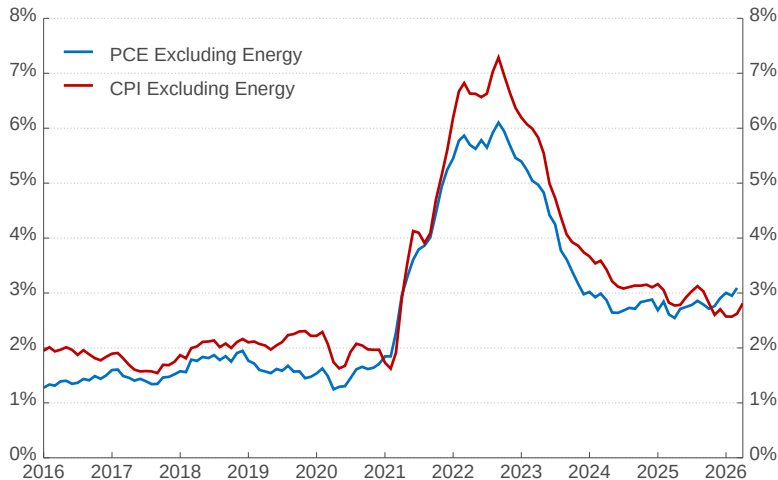
# CURRENT INFLATION IN HISTORICAL CONTEXT

PCE EXCLUDING ENERGY ANNUAL INFLATION [▷ BACK](#)



# STALLED PROGRESS ON PCE INFLATION WHEN EXCLUDING ENERGY

PCE AND CPI EXCLUDING ENERGY SHOWS LACK OF PROGRESS ON ANNUAL INFLATION [▷ BACK](#)





# CPI INFLATION AGGREGATES

PROGRESS PERSISTED LONGER FOR CPI BUT IT HAS ALSO STALLED [▷ BACK](#)

## Annualized CPI Inflation Rates

|          | All Goods<br>& Services | Excluding<br>Energy | Excluding Food<br>& Energy (Core) |
|----------|-------------------------|---------------------|-----------------------------------|
| 2016–19  | 2.1%                    | 2.0%                | 2.1%                              |
| 2020     | 1.3%                    | 2.0%                | 1.6%                              |
| 2021     | 7.2%                    | 5.6%                | 5.5%                              |
| 2022     | 6.4%                    | 6.4%                | 5.7%                              |
| 2023     | 3.3%                    | 3.7%                | 3.9%                              |
| 2024     | 2.9%                    | 3.1%                | 3.2%                              |
| 2025     | 2.7%                    | 2.7%                | 2.6%                              |
| 2026 YTD | 6.0%                    | 3.3%                | 3.3%                              |

SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is through April; the swaps column remains a manual input.

NOTE: The 2026 period is through April; the swaps column remains a manual input.

NOTE: The

2026 period is the year through April.

# CPI INFLATION COMPONENTS

REVERSALS IN FOOD, ENERGY AND CORE GOODS; MODERATION IN CORE SERVICES AND HOUSING [▷ BACK](#)

## Annualized CPI Inflation Rates by Categories

|          | Food  | Energy | Core<br>Commodities | Services<br>Less Rent | Rent of<br>Shelter |
|----------|-------|--------|---------------------|-----------------------|--------------------|
| 2016–19  | 1.2%  | 3.9%   | −0.2%               | 2.2%                  | 3.3%               |
| 2020     | 3.9%  | −7.0%  | 1.6%                | 1.4%                  | 1.9%               |
| 2021     | 6.0%  | 30.3%  | 10.7%               | 3.9%                  | 4.2%               |
| 2022     | 10.1% | 6.9%   | 2.1%                | 7.5%                  | 7.6%               |
| 2023     | 2.7%  | −1.9%  | 0.1%                | 3.5%                  | 6.2%               |
| 2024     | 2.4%  | −0.3%  | −0.7%               | 4.0%                  | 4.6%               |
| 2025     | 3.0%  | 2.1%   | 1.4%                | 3.4%                  | 3.1%               |
| 2026 YTD | 3.2%  | 48.7%  | 0.8%                | 4.1%                  | 4.0%               |

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through April.

# HIGH CORE INFLATION IS WIDESPREAD

ESTIMATED DISTRIBUTION OF CORE PCE INFLATION [▷ BACK](#)

